Chapter 3

Newton's Laws of Motion

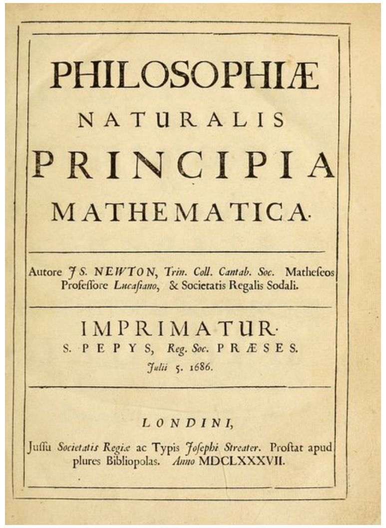
Introduction

- We've seen how to use ***kinematics*** to describe motion in one, two, or three dimensions.
- But what *causes* bodies to move the way that they do?
- The answer takes us into the subject of ***dynamics***, the relationship of motion to the forces that cause it.

Introduction

- The principles of dynamics were clearly stated for the first time by Sir Isaac Newton in his book "Mathematical Principles of Natural Philosophy" (1687); today we call them Newton's laws of motion.

- Newton did not *derive* these laws, but rather deduced them from a multitude of experiments performed by other scientists, including Galileo and Huygens.



PHILOSOPHIÆ
NATURALIS
PRINCIPIA
MATHEMATICA.

Autore *J S.* NEWTON, *Trin. Coll. Cantab. Soc.* Mathefeos Profeffore *Lucafiano*, & Societatis Regalis Sodali.

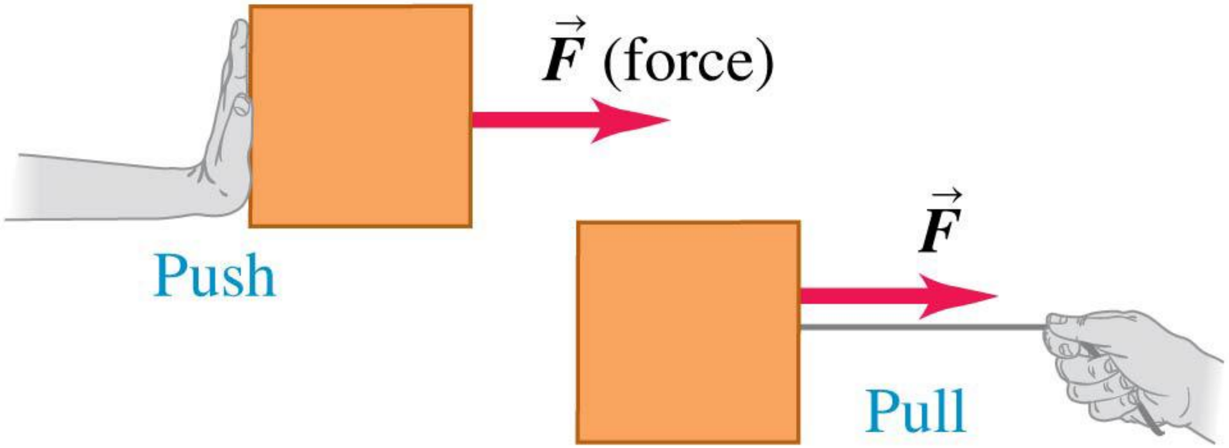
IMPRIMATUR.
S. PEPYS, *Reg. Soc.* PRÆSES.
Julii 5. 1686.

LONDINI,

Juffu *Societatis Regiæ* ac Typis *Jofephi Streater.* Proftat apud plures Bibliopolas. *Anno* MDCLXXXVII.

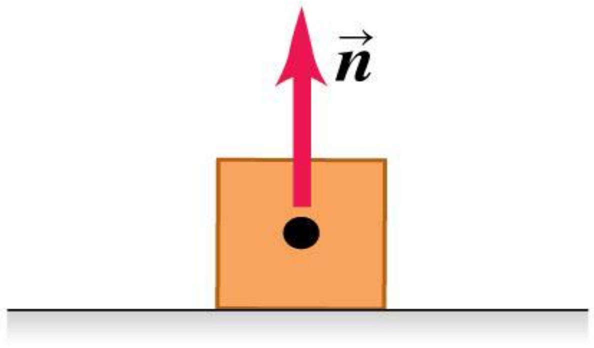
What are some properties of a force?

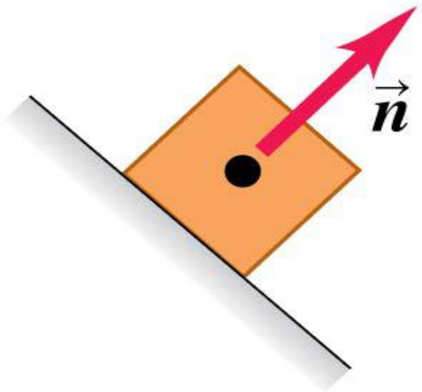
- A force is a push or a pull.
- A force is an interaction between two objects or between an object and its environment.
- A force is a vector quantity, with magnitude and direction.



Common types of forces: Normal force

Normal force $\vec{n}$: When an object rests or pushes on a surface, the surface exerts a push on it that is directed perpendicular to the surface.

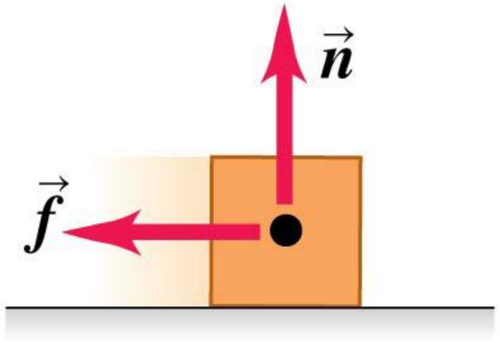




- The normal force is a contact force.

Common types of forces: Friction

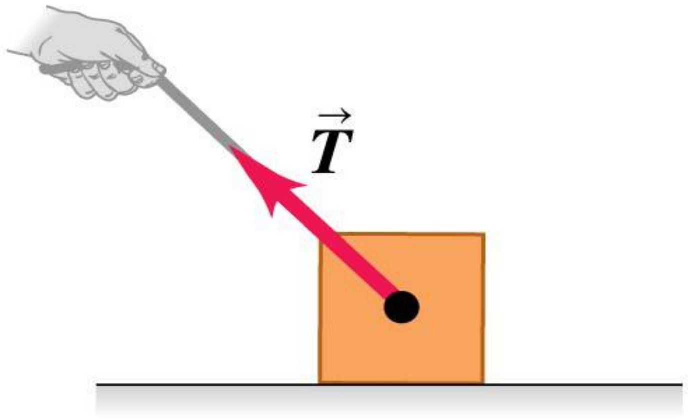
Friction force $\vec{f}$: In addition to the normal force, a surface may exert a friction force on an object, directed parallel to the surface.



- Friction is a contact force.

Common types of forces: Tension

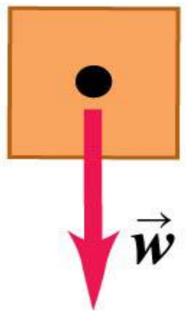
Tension force $\vec{T}$: A pulling force exerted on an object by a rope, cord, etc.



- Tension is a contact force.

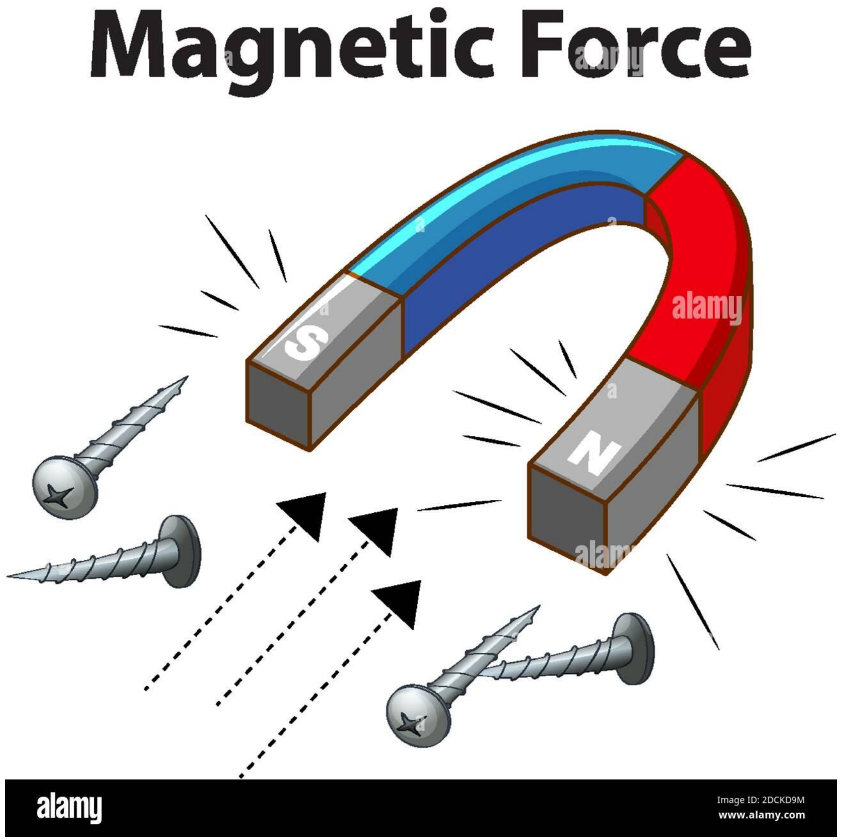
Common types of forces: Weight

Weight $\vec{w}$: The pull of gravity on an object is a long-range force (a force that acts over a distance).



- Weight is a long-range force.

Common types of forces: Electric and Magnetic force



Electric and Magnetic forces are long-range forces.

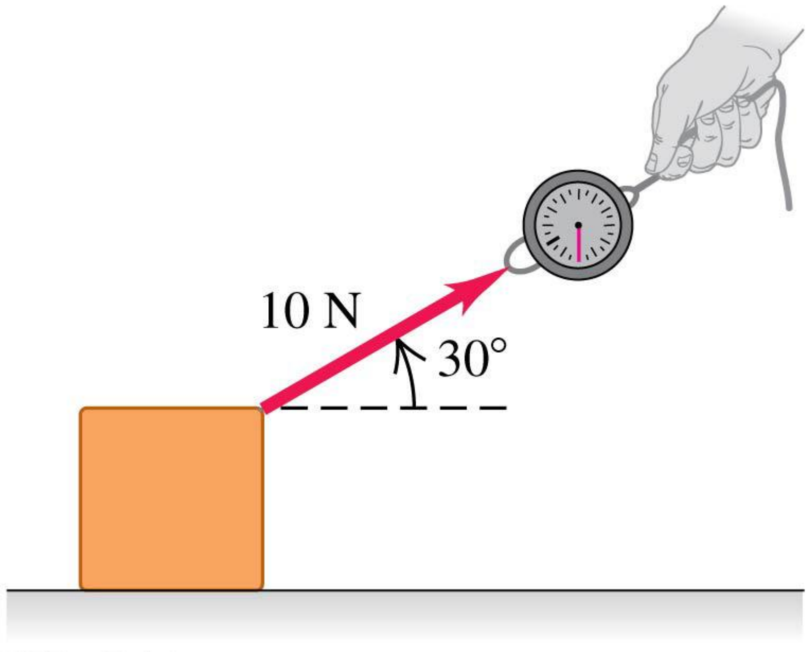
What are the magnitudes of common forces?

- The SI unit of the magnitude of force is the newton, abbreviated N. Some typical force magnitudes are:

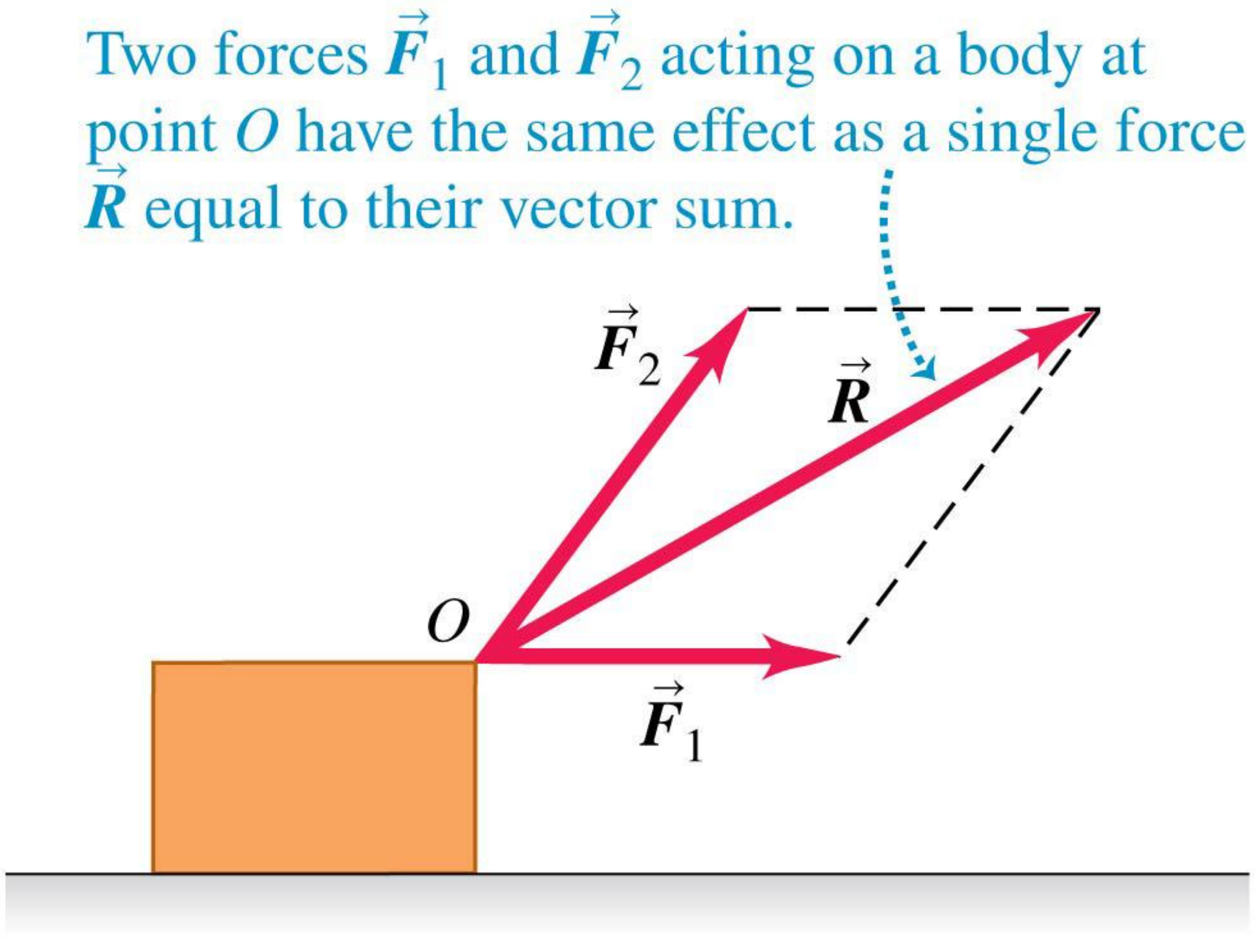
Weight of a large blue whale	1.9×10^{6} N
Maximum pulling force of a locomotive	8.9×10^{5} N
Weight of a 250-lb linebacker	1.1×10^{3} N
Weight of a medium apple	1 N
Electric attraction between the proton and the electron in a hydrogen atom	8.2×10^{-8} N
Gravitational attraction between the proton and the electron in a hydrogen atom	3.6×10^{-47} N

Drawing force vectors

- The figure shows a spring balance being used to measure a pull that we apply to a box.
- We draw a vector to represent the applied force.
- The length of the vector shows the magnitude; the longer the vector, the greater the force magnitude.

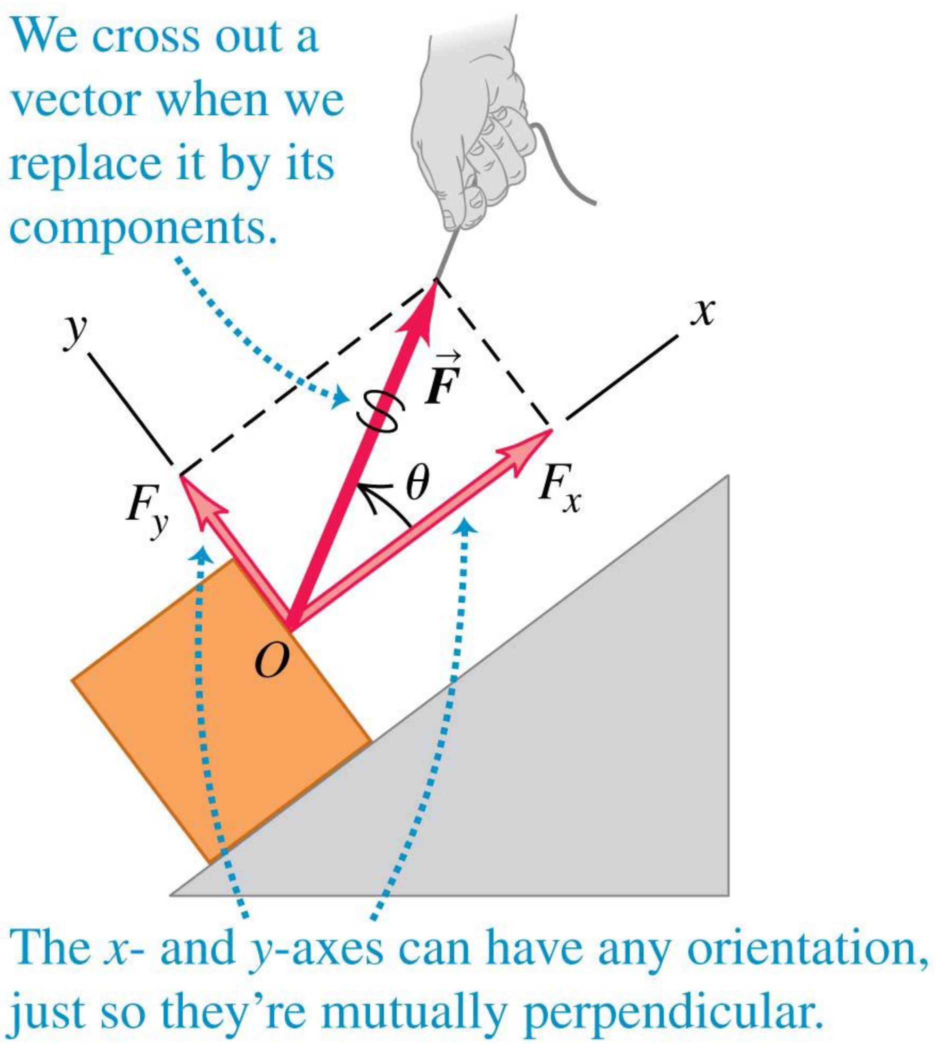


Superposition of forces



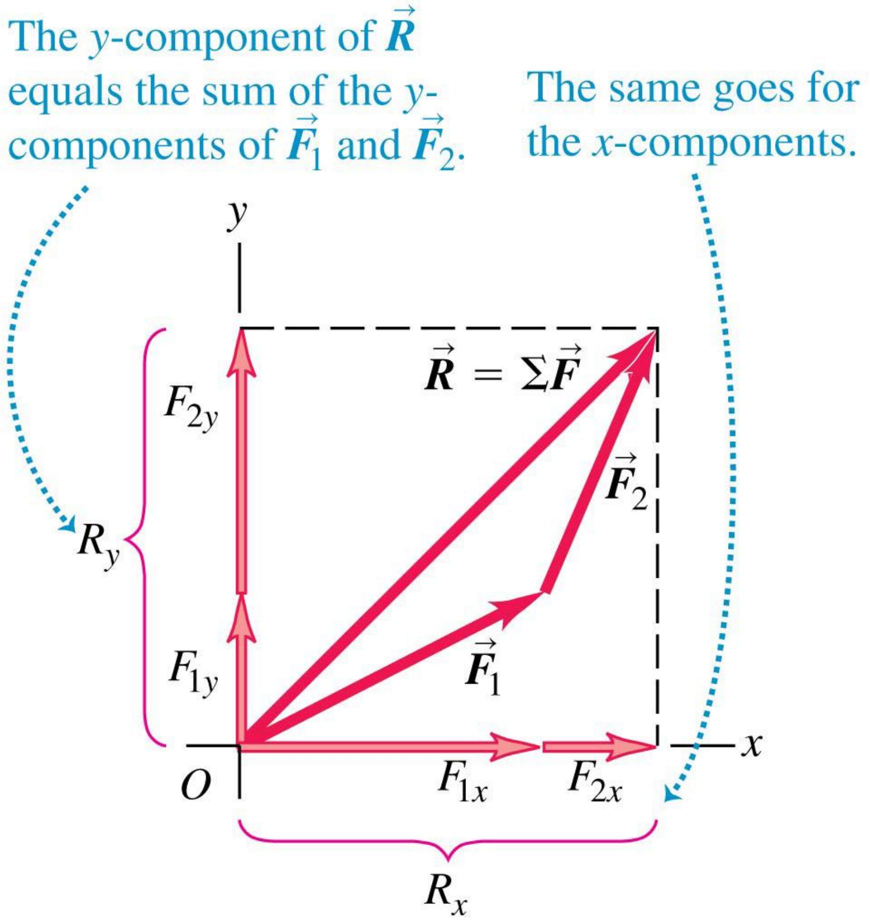
- Several forces acting at a point on an object have the same effect as their vector sum acting at the same point.

Decomposing a force into its component vectors



- Choose perpendicular x- and y-axes.
- F_x and F_y are the components of a force along these axes.
- Use trigonometry to find these force components.

Notation for the vector sum



- The vector sum of all the forces on an object is called the **resultant** of the forces or the **net force**:

$$\vec{R} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \cdots = \Sigma\vec{F}$$

Newton’s first law

Newton’s laws of motion

First Law When all external influences on a particle are removed, the particle moves with constant velocity. [This velocity may be zero in which case the particle remains at rest.]

In which reference frame is Newton's first law valid?

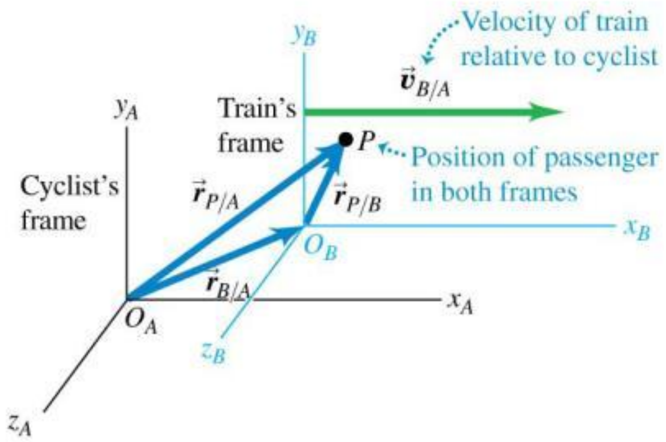
- Suppose you are in a bus that is traveling on a straight road and speeding up.
- If you could stand in the aisle on roller skates, you would start moving *backward* relative to the bus as the bus gains speed.
- It looks as though Newton's first law is not obeyed; there is no net force acting on you, yet your velocity changes.
- The bus is accelerating with respect to the earth and is not a suitable frame of reference for Newton's first law.

In which reference frame is Newton's first law valid?

- Because of the influence of the Earth's gravity, any experiment conducted on the Earth's surface cannot verify the First Law.
- All the external influences must be removed.
- We can think about conducting an experiment in a place as remote as possible from any galaxies and material bodies.

In which reference frame is Newton's first law valid?

- In the universe, everything is moving relative to each other.
- If the First Law has been found to be true in reference frame F, then it should be true in any other frame F' that is mutually unaccelerated relative to F.
- The First Law does not hold in any reference frame that is accelerated relative to F.



Inertial frame of reference

- Definition of Inertial frame:
  A reference frame in which the First Law is true is said to be an inertial frame.

- If there exists one inertial frame, then there exist infinitely many frames (any frame moving with constant velocity relative to the one) that the First Law holds.

- That sounds like a circular argument! Is there any physical content? In principle, our Universe can have no inertial frame at all. But in reality, the inertial frames do exist in nature. This fact is called the law of inertia.

- The law of inertia: There exists in nature a unique class of mutually unaccelerated reference frames (the inertial frames) in which the First Law is true.

Approximate (not exact) inertial frame

The Earth's rotation is not large and its orbital acceleration is small.

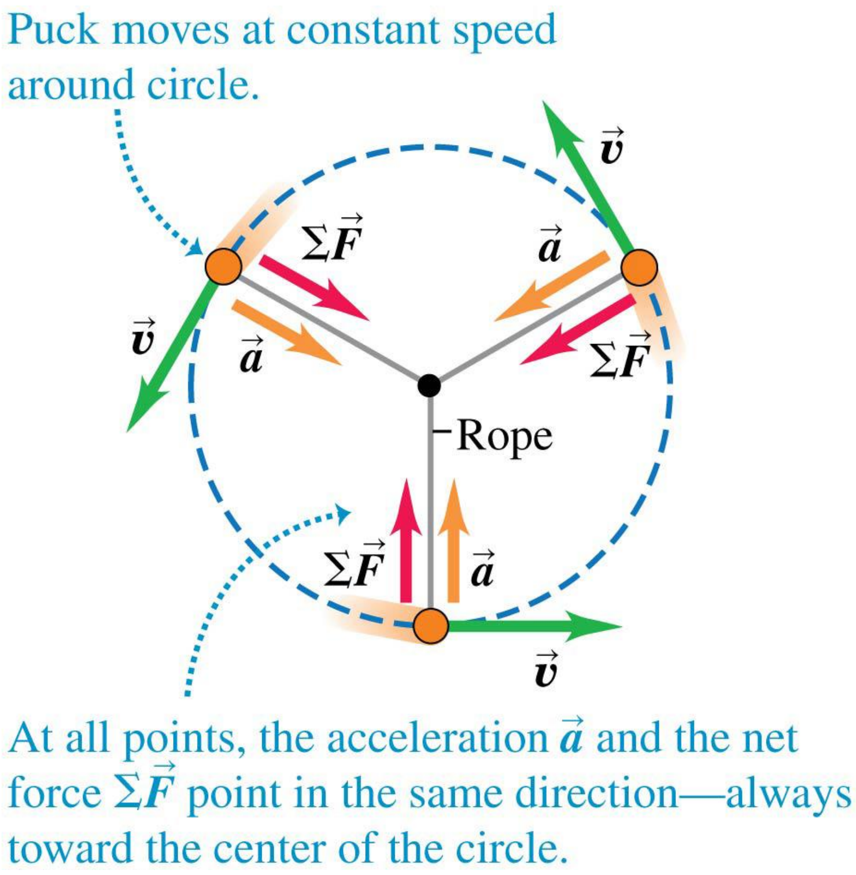
For problems concerned with motions at or near the earth's surface, we typically assume our "inertial frame" to be fixed to the earth. We neglect any acceleration effects from the earth's rotation or its orbital acceleration.

For problems involving satellites or rockets, the inertial frame of reference is often fixed to the Sun or other stars.

Non-Inertial frame of reference

- As we have already seen, an object in uniform circular motion is accelerated toward the center of the circle. So the net force on the object must point toward the center of the circle.

- The frame following the object is also a non-inertial frame.



Quiz

Suppose that a reference frame fixed to the Earth is exactly inertial. Which of the following are then inertial frames? A frame fixed to a motor car which is

(i) moving with constant speed around a flat race track,

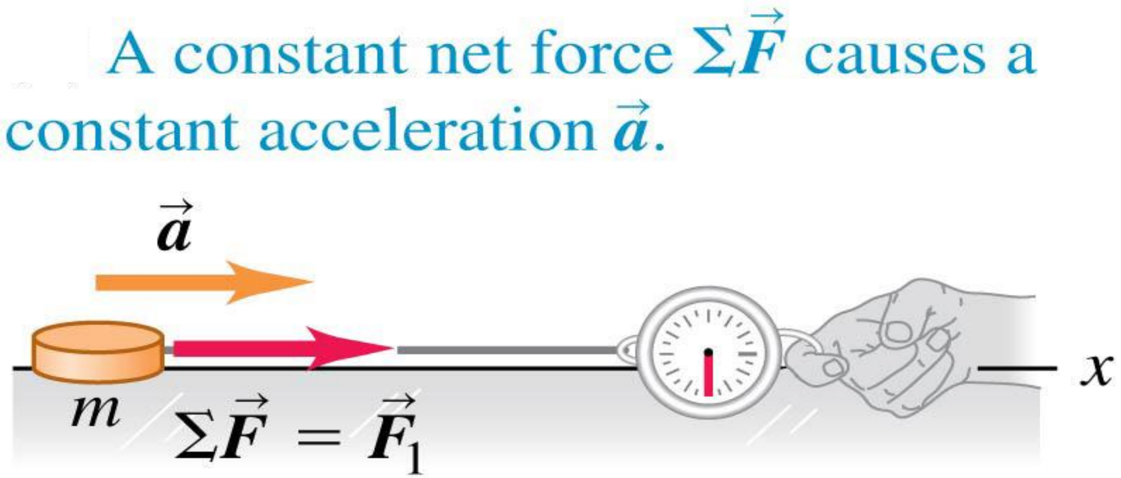
(ii) moving with constant speed along a straight undulating road,

(iii) moving with constant speed up a constant gradient,

(iv) freewheeling down a hill.

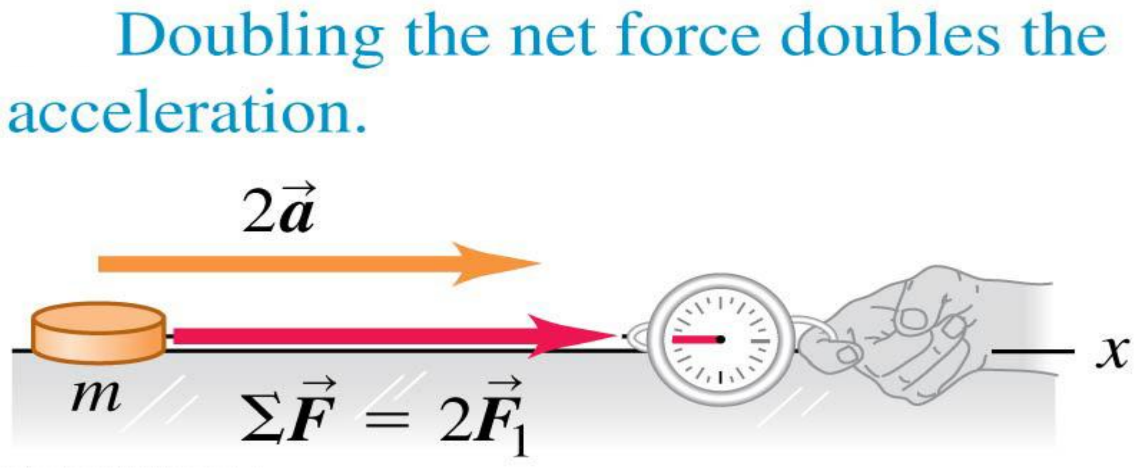
Force and acceleration

- The acceleration $\vec{a}$ of an object is directly proportional to the net force $\Sigma\vec{F}$ on the object.



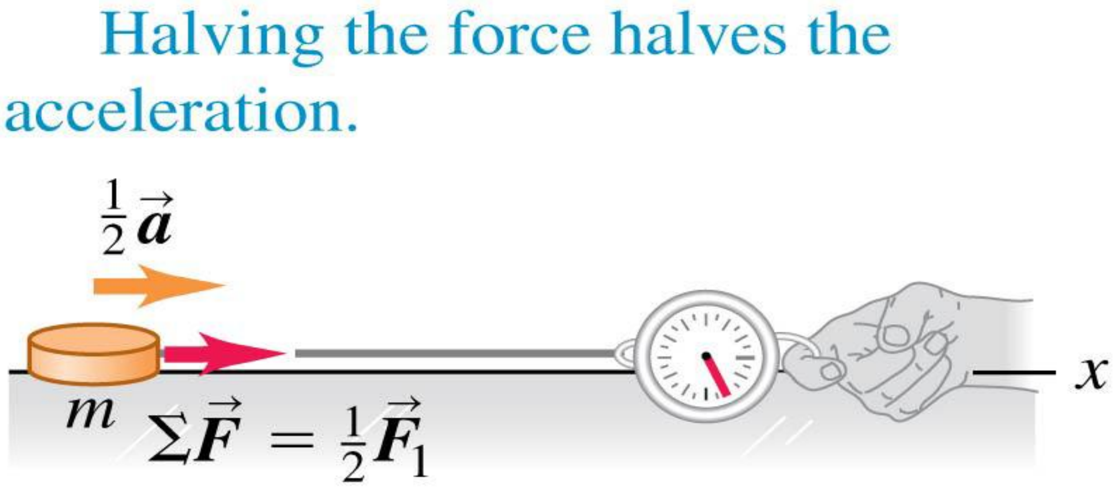
Force and acceleration

- The acceleration $\vec{a}$ of an object is directly proportional to the net force $\Sigma\vec{F}$ on the object.



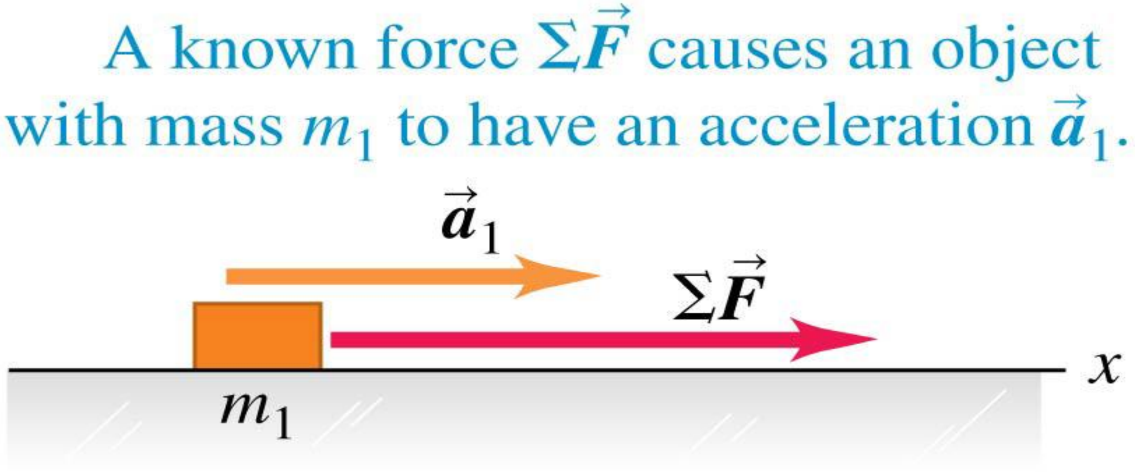
Force and acceleration

- The acceleration $\vec{a}$ of an object is directly proportional to the net force $\Sigma\vec{F}$ on the object.



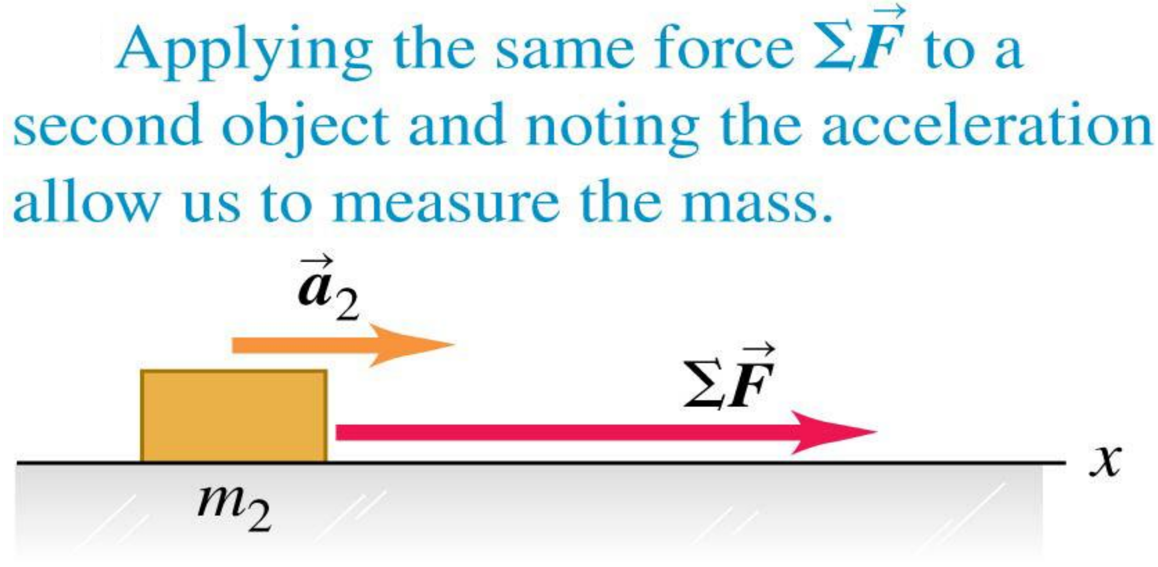
Mass and acceleration

- The acceleration of an object is inversely proportional to the object's mass if the net force remains fixed.



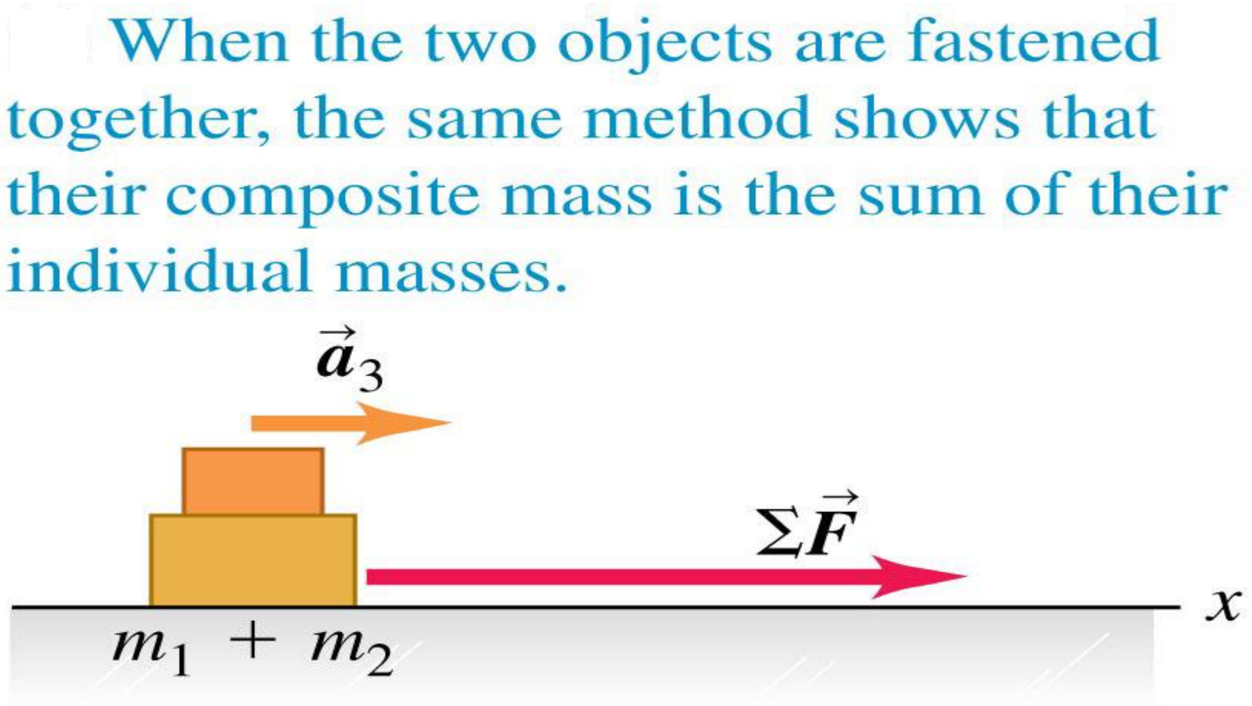
Mass and acceleration

- The acceleration of an object is inversely proportional to the object's mass if the net force remains fixed.



Mass and acceleration

- The acceleration of an object is inversely proportional to the object's mass if the net force remains fixed.



Newton's second law of motion

Second Law When a force $\boldsymbol{F}$ acts on a particle of mass m, the particle moves with instantaneous acceleration $\boldsymbol{a}$ given by the formula

$$\boldsymbol{F} = m\boldsymbol{a},$$

- The SI unit for force is the newton (N).

$$1 \text{ N} = 1 \text{ kg} \cdot \text{m/s}^2$$

Mass and weight

- It is important to understand the difference between the mass and weight of a body!

- Mass is an absolute property of a body. It is independent of the gravitational field in which it is measured. The mass provides a measure of the resistance of a body to a change in velocity, as defined by Newton's second law of motion ($m = \vec{F}/\vec{a}$).

- We call such mass the inertial mass.

Mass and weight

- The *weight* of an object (on the earth) is the gravitational force that the earth exerts on it.
- The weight of a body is not absolute, since it depends on the gravitational field in which it is measured.
- The weight w of an object of mass m^* is

$$w = m^* g$$

where g is the gravitational acceleration constant.

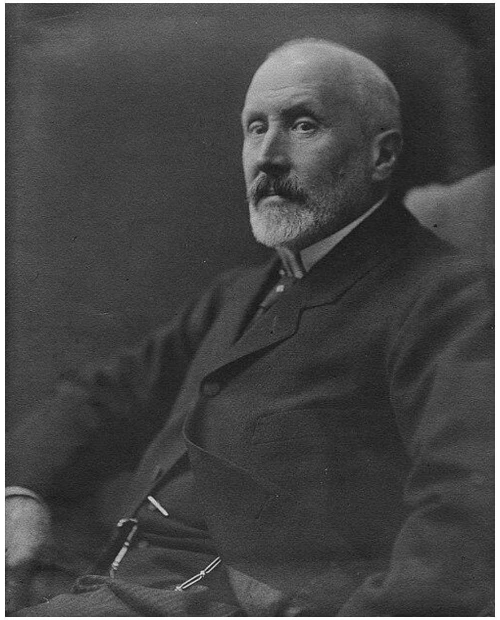
- We call m* the gravitational mass.

The equivalence principle

In principle, the gravitational mass has nothing to do with the inertial mass. A particle can have two "masses": inertial mass m and gravitational mass m^*.

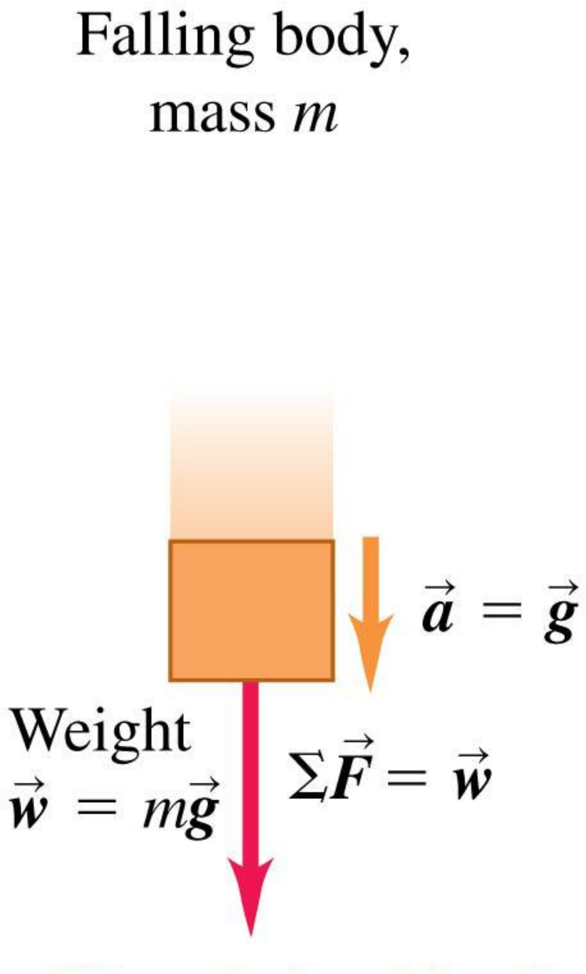
However, experiments of Eötvös (1890) and later refinements found the difference between m and m* to be less than 10^{-11}.

Therefore, we believe the inertial and gravitational mass are exactly equal, $m=m^*$. This is called the principle of equivalence.



Loránd Eötvös

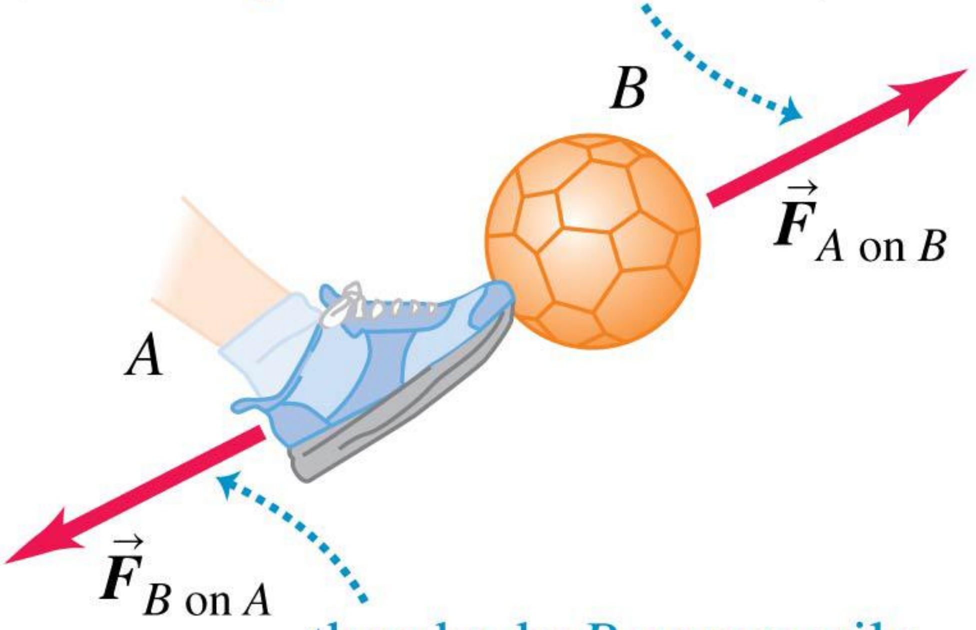
Relating the mass and weight of a body



The gravitational acceleration g (at the same place) is the same for any particle.

Newton’s third law

If body A exerts force $\vec{F}_{A \text{ on } B}$ on body B
(for example, a foot kicks a ball) ...

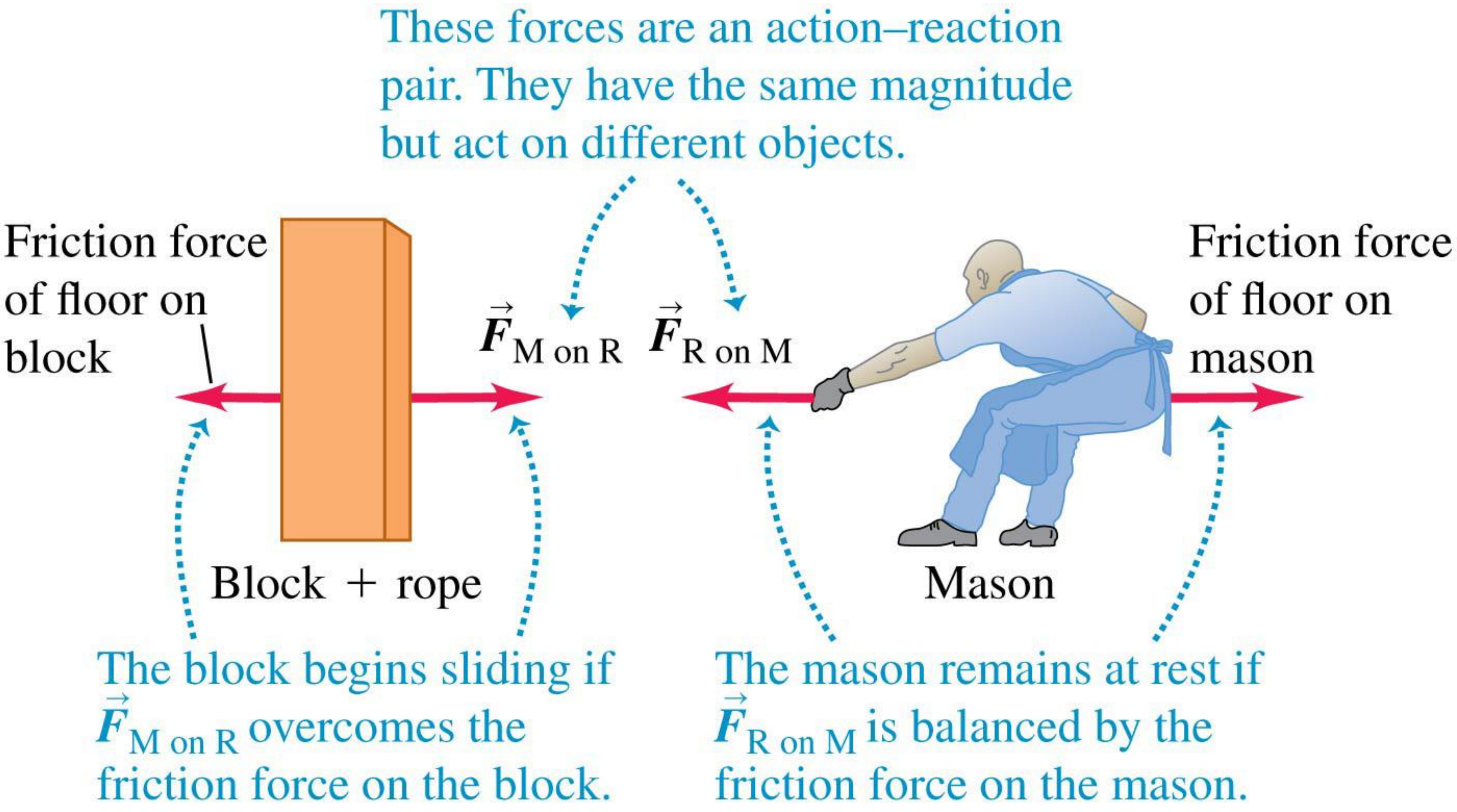


... then body B necessarily
exerts force $\vec{F}_{B \text{ on } A}$ on body A
(ball kicks back on foot).

The two forces have same magnitude
but opposite directions: $\vec{F}_{A \text{ on } B} = -\vec{F}_{B \text{ on } A}$.

A paradox?

- If an object pulls back on you just as hard as you pull on it, how can it ever accelerate?



Newton’s third law

Third Law When two particles exert forces upon each other, these forces are (i) equal in magnitude, (ii) opposite in direction, and (iii) parallel to the straight line joining the two particles.

Quiz

Two people pull on opposite ends of a rope, each with a force F. The tension in the rope is

A. 2F
B. F
C. F/2
D. 0

Newton's law of motion: summary

Newton's laws of motion

First Law When all external influences on a particle are removed, the particle moves with constant velocity. [This velocity may be zero in which case the particle remains at rest.]

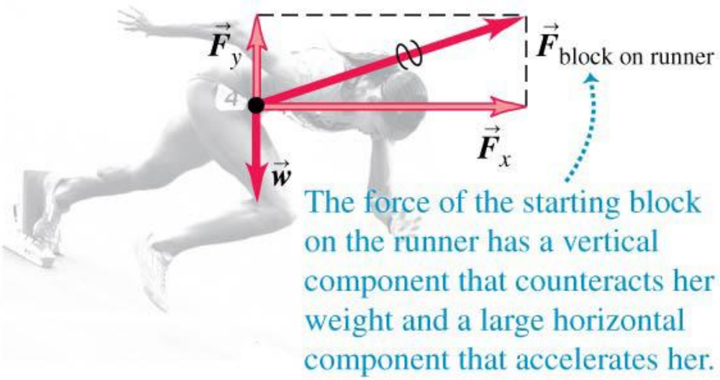
Second Law When a force $\boldsymbol{F}$ acts on a particle of mass m, the particle moves with instantaneous acceleration $\boldsymbol{a}$ given by the formula

$$\boldsymbol{F} = m\boldsymbol{a},$$

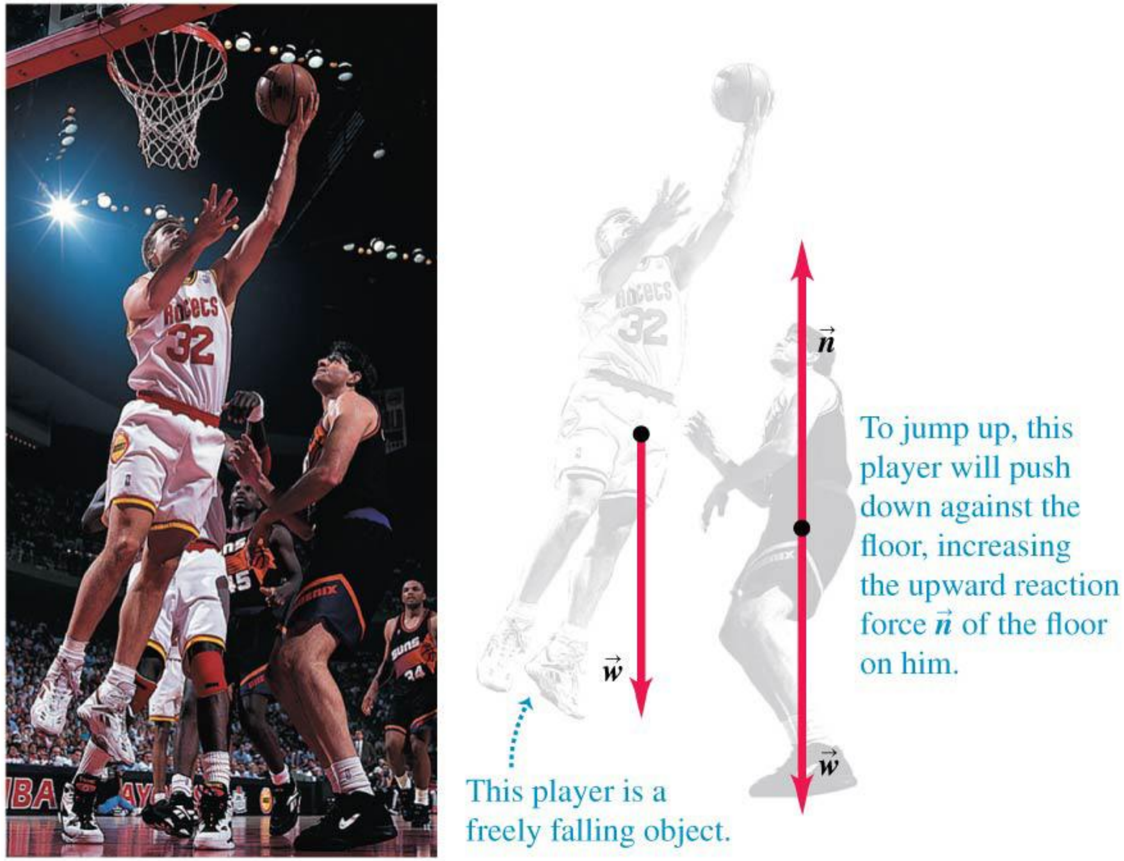
where the unit of force is implied by the units of mass and acceleration.

Third Law When two particles exert forces upon each other, these forces are (i) equal in magnitude, (ii) opposite in direction, and (iii) parallel to the straight line joining the two particles.

Free-body diagrams

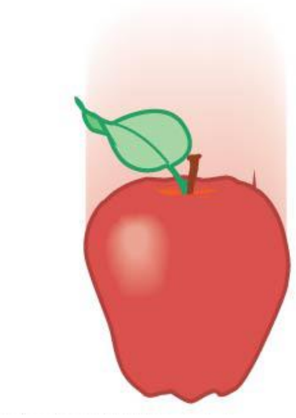


Free-body diagrams



A note on free-body diagrams

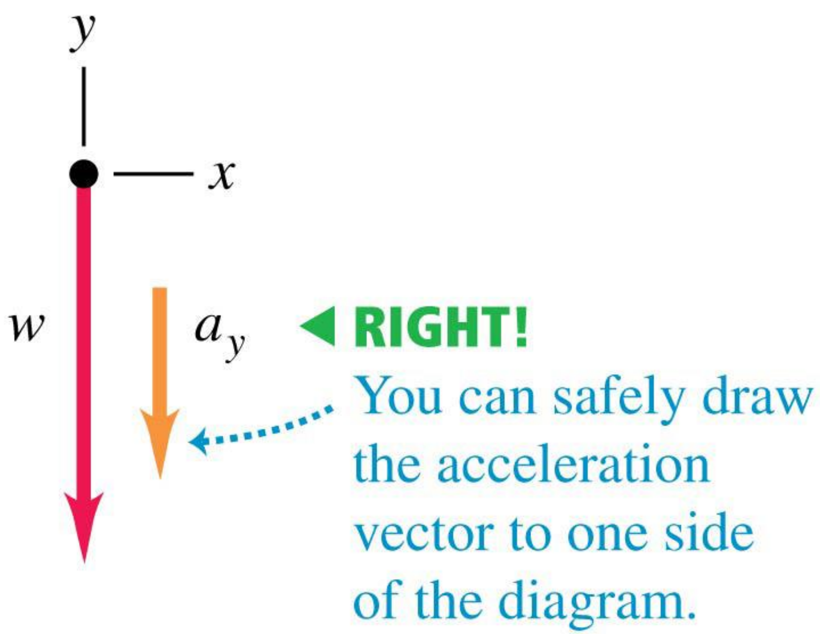
- $m\vec{a}$ does *not* belong in a free-body diagram.



Only the force of gravity acts on this falling fruit.

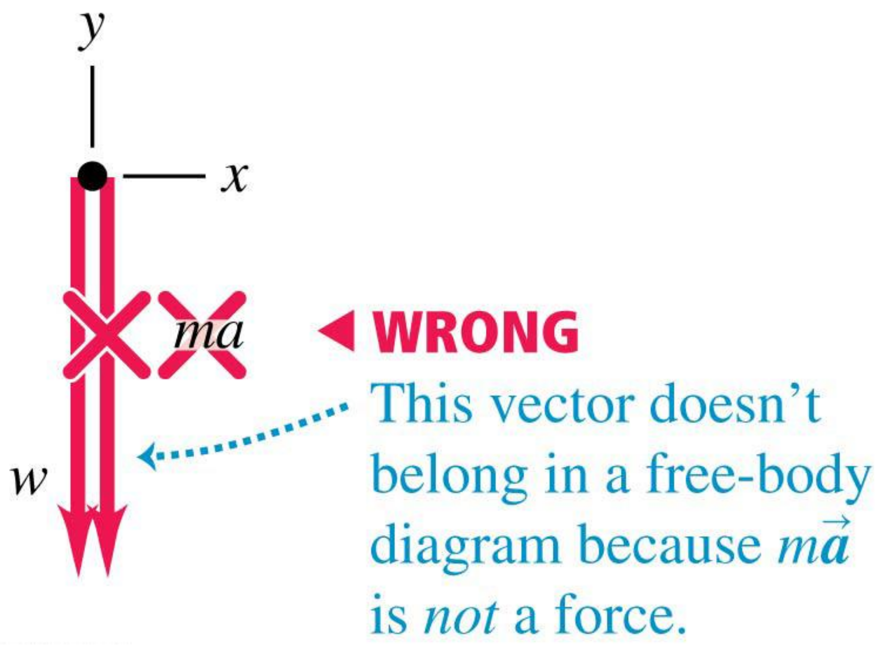
A note on free-body diagrams

- Correct free-body diagram

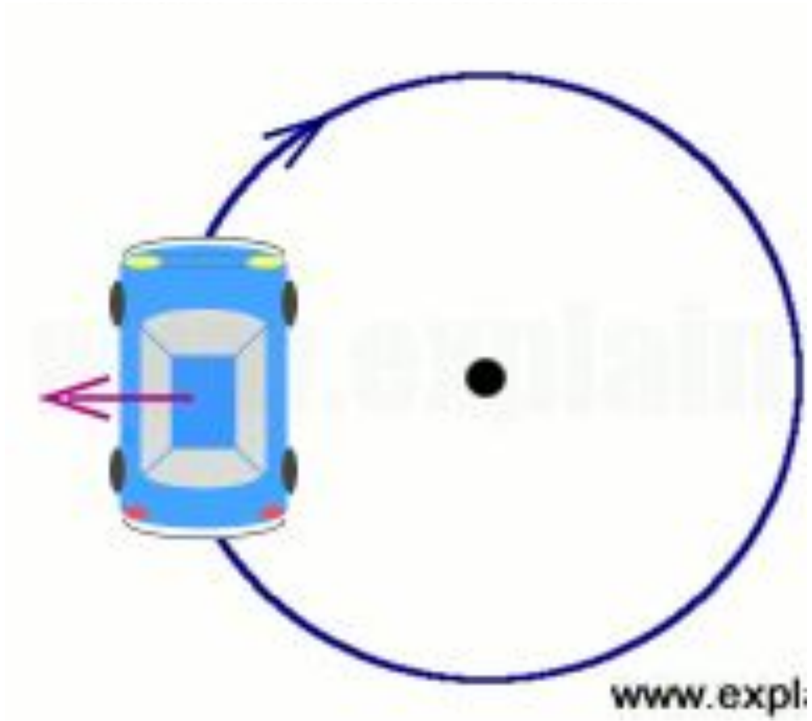


A note on free-body diagrams

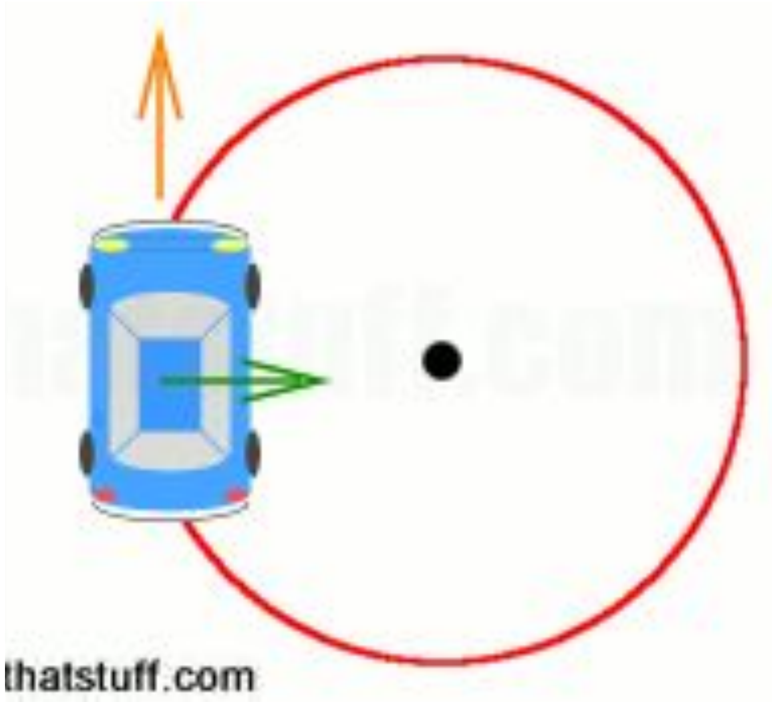
- Incorrect free-body diagram



A note on “centrifugal force”

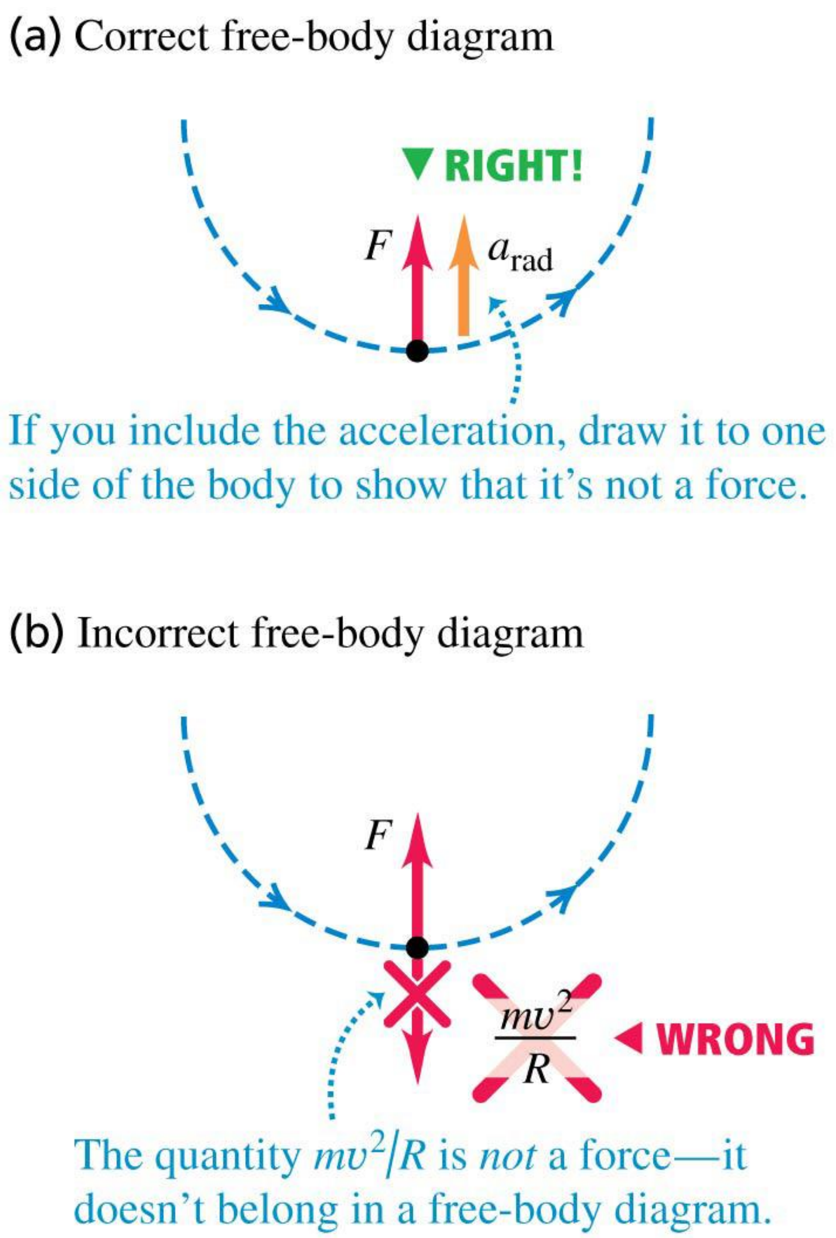


Centrifugal force is fictitious

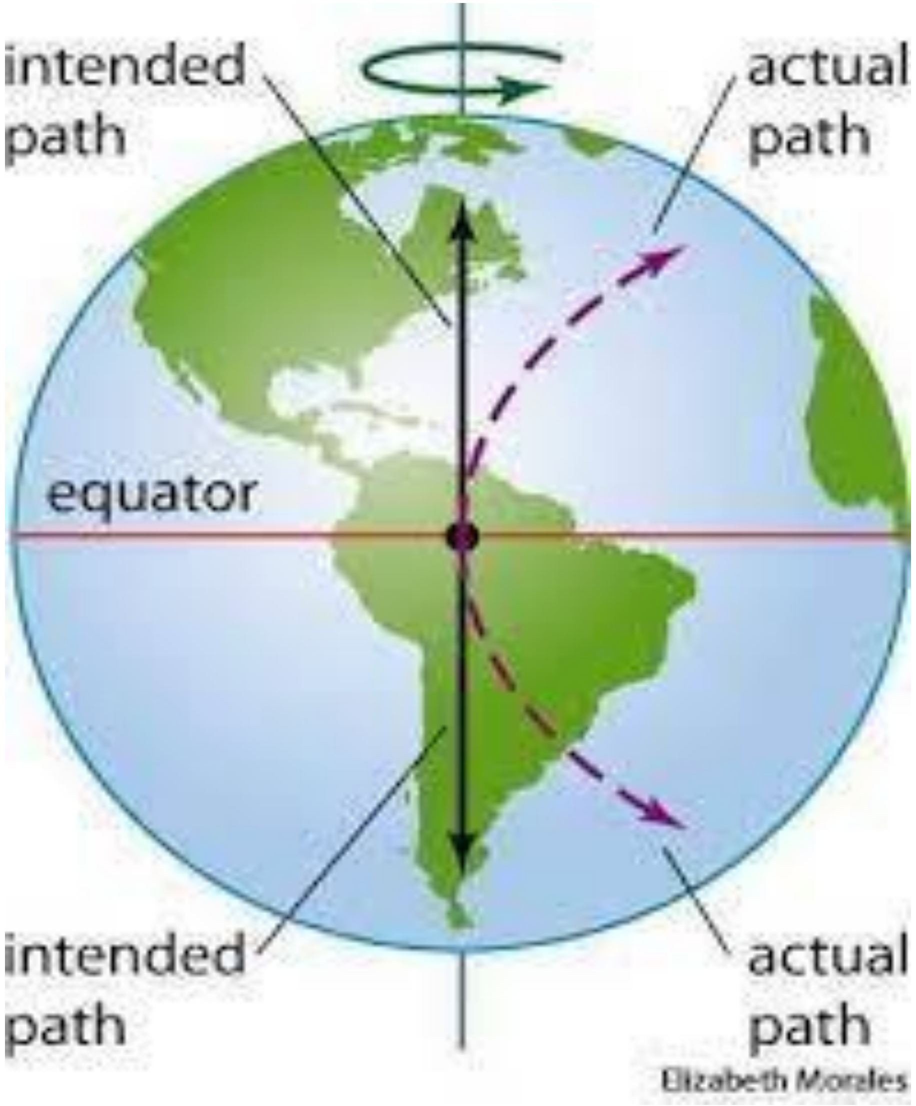


Avoid using "centrifugal force"

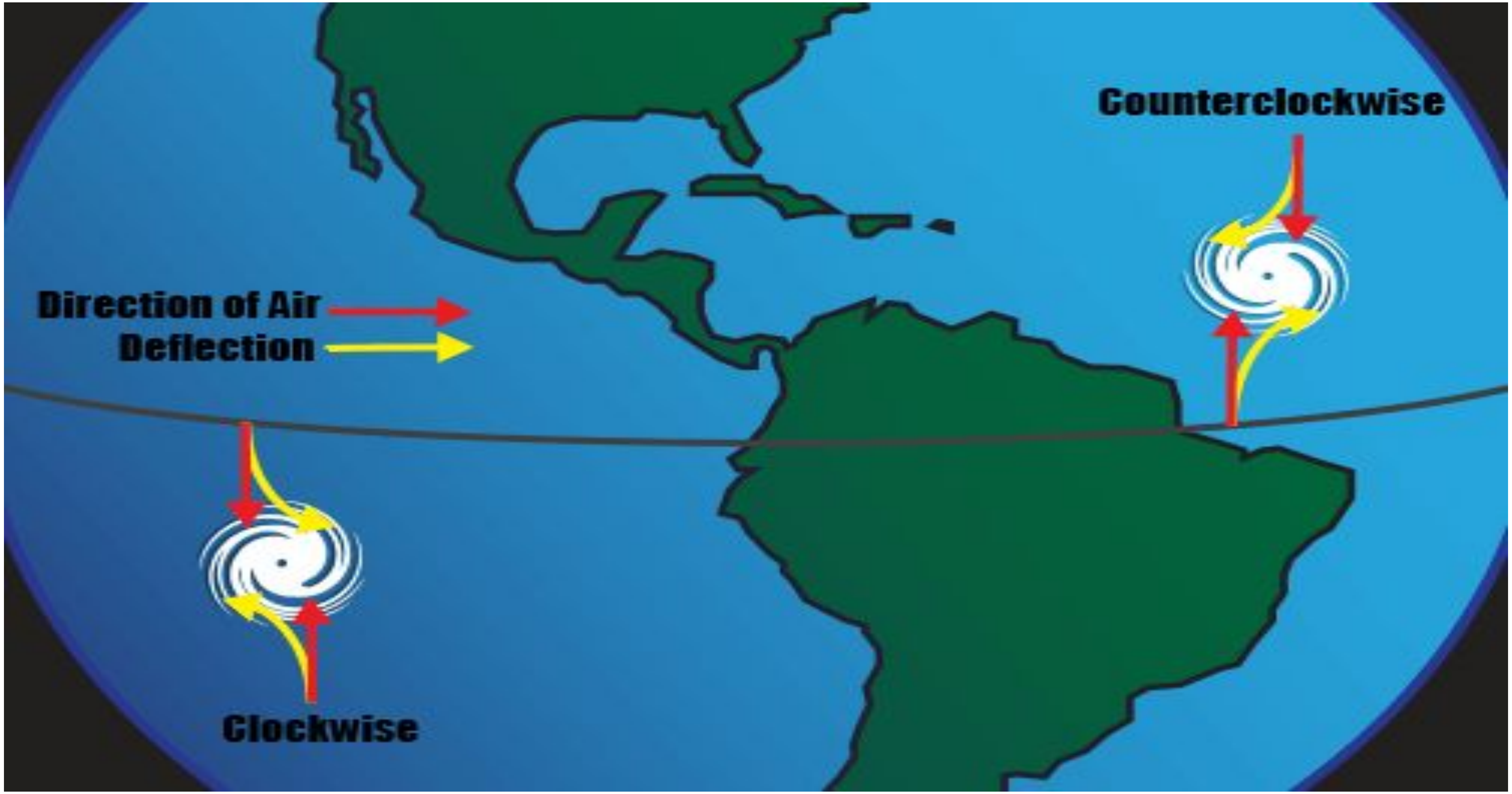
- Figure (a) shows the correct free-body diagram for a body in uniform circular motion.
- Figure (b) shows a common error.
- In an inertial frame of reference, there is no such thing as "centrifugal force."



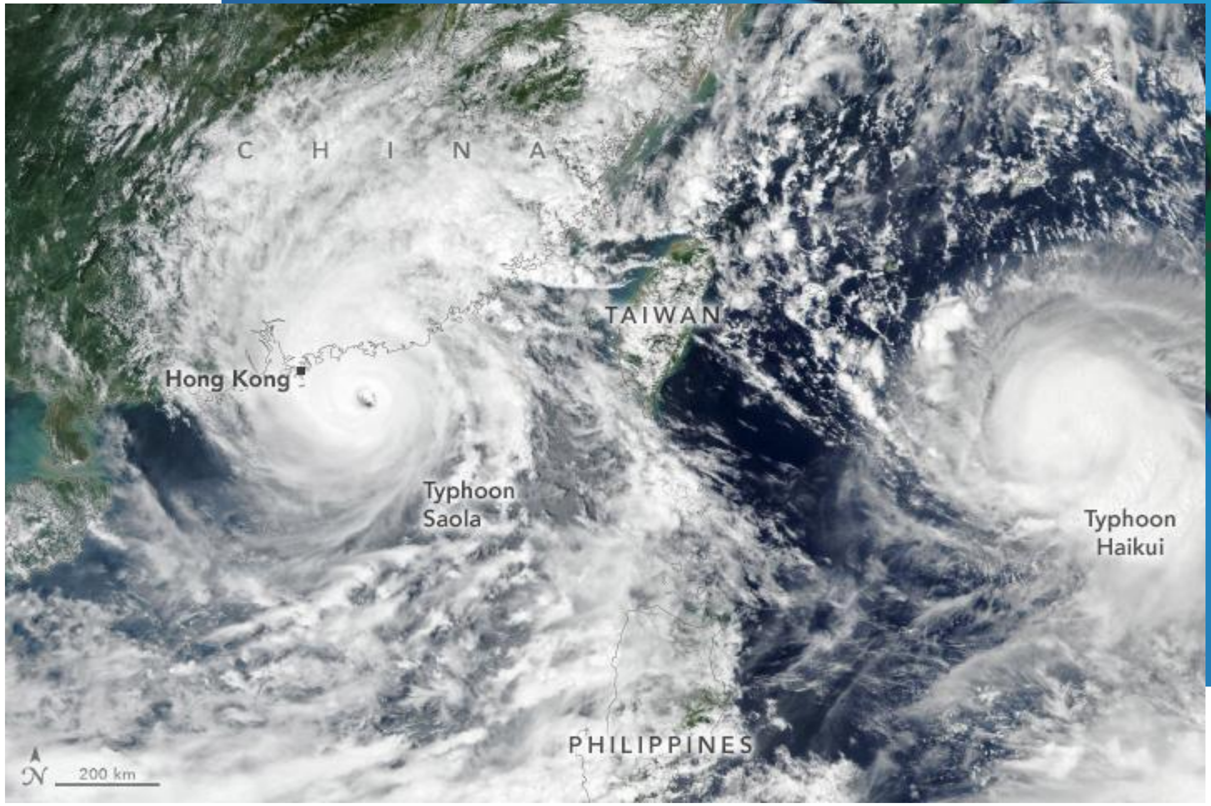
Coriolis force: another famous fictitious force



Coriolis force: another famous fictitious force



Coriolis force: another famous fictitious force



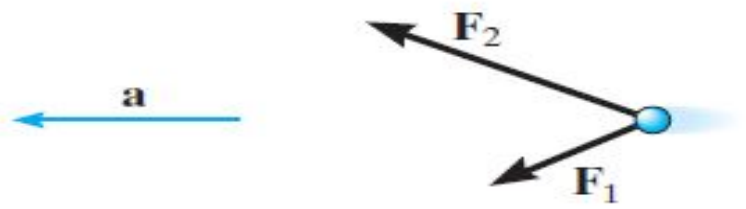
Typhoon Saola

Equation of motion

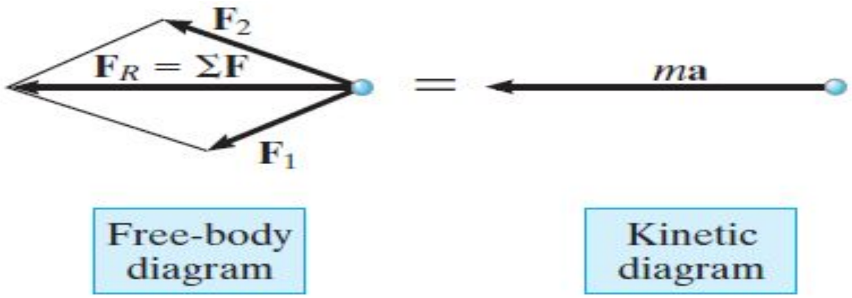
The motion of a particle is governed by Newton's second law, relating the unbalanced forces on a particle to its acceleration. If more than one force acts on the particle, the equation of motion can be written

$$\sum \vec{\boldsymbol{F}} = \vec{\boldsymbol{F}}_{\mathrm{R}} = \mathrm{m}\,\vec{\boldsymbol{a}}$$

where $\vec{\boldsymbol{F}}_{\mathrm{R}}$ is the resultant force, which is a vector summation of all the forces.



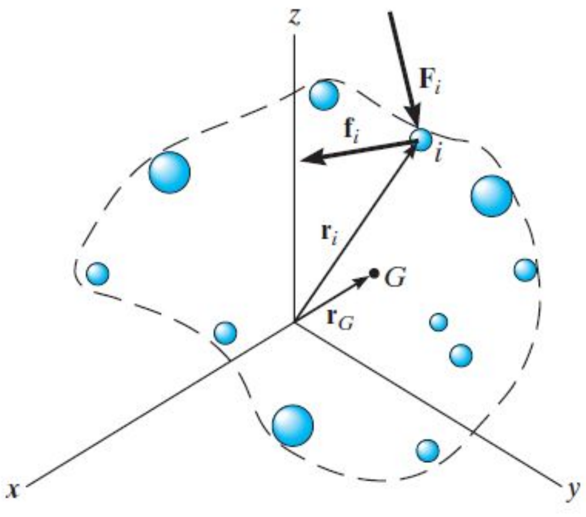
To illustrate the equation, consider a particle acted on by two forces.



First, draw the particle's free-body diagram, showing all forces acting on the particle. Next, draw the kinetic diagram, showing the inertial force m**a** acting in the same direction as the resultant force $\boldsymbol{F}_{\mathrm{R}}$.

Equation of motion for a system of particles

The equation of motion can be extended to include systems of particles. This includes the motion of solids, liquids, or gas systems.



Inertial coordinate system

As in statics, there are internal forces and external forces acting on the system. What is the difference between them?

Using the definitions of $m = \sum m_i$ as the total mass of all particles and $\vec{\mathbf{a}}_G$ as the acceleration of the center of mass G of the particles, then $m\,\vec{\mathbf{a}}_G = \sum m_i\,\vec{\mathbf{a}}_i$.

For a system of particles: $\sum \vec{\mathbf{F}} = m\,\vec{\mathbf{a}}_G$ where $\sum \vec{\mathbf{F}}$ is the sum of the external forces acting on the entire system.

Key points

1) Newton's second law is a "law of nature"-- experimentally proven, not the result of an analytical proof.

2) Mass (property of an object) is a measure of the resistance to a change in velocity of the object.

3) Weight (a force) depends on the local gravitational field. Calculating the weight of an object is an application of
$F = m\ a$, i.e., $w = m\ g$.

4) Unbalanced forces cause the acceleration of objects. This condition is fundamental to all dynamics problems!

Procedure for the application of the equation of motion

1) Select a convenient coordinate system. Rectangular, normal/tangential, or cylindrical coordinates may be used.

2) Draw a free-body diagram showing all external forces applied to the particle. Resolve forces into their appropriate components.

3) Draw the kinetic diagram, showing the particle's inertial force, m $\vec{a}$. Resolve this vector into its appropriate components.

4) Apply the equations of motion in their scalar component form and solve these equations for the unknowns.

5) It may be necessary to apply the proper kinematic relations to generate additional equations.

Quiz

1. Newton's second law can be written in mathematical form as $\Sigma\ \vec{\mathbf{F}} = m\ \vec{\mathbf{a}}$. Within the summation of forces, $\Sigma\vec{\mathbf{F}}$, __________ are(is) not included.

   A) external forces B) weight

   C) internal forces D) All of the above.

2. The equation of motion for a system of n-particles can be written as $\Sigma\ \vec{\mathbf{F}}_i = \Sigma\ m_i\ \vec{\mathbf{a}}_i = m\ \vec{\mathbf{a}}_G$, where $\vec{\mathbf{a}}_G$ indicates _________.

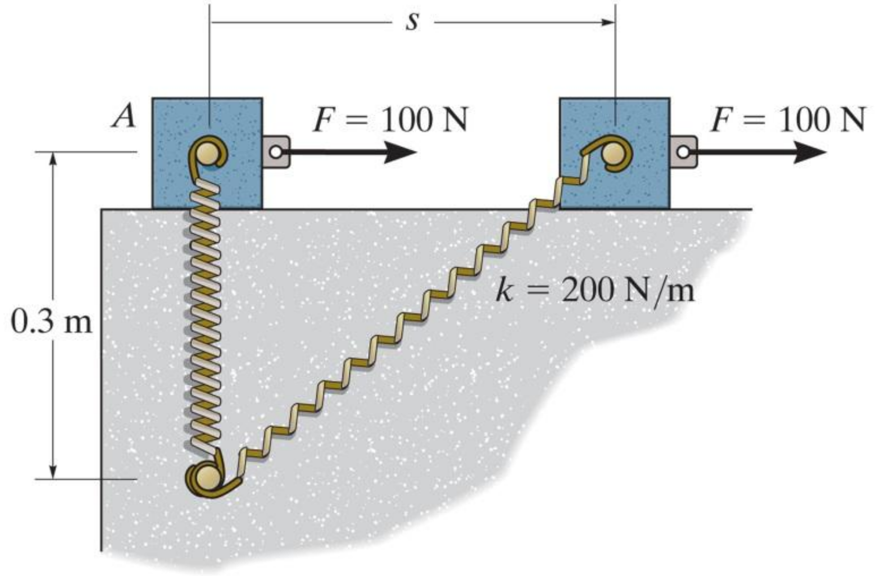
   A) summation of each particle's acceleration

   B) acceleration of the center of mass of the system

   C) acceleration of the largest particle

   D) None of the above.

Example 1



Given: A 25-kg block is subjected to the force F=100 N. The spring has a stiffness of k = 200 N/m and is un-stretched when the block is at A. The contact surface is smooth.

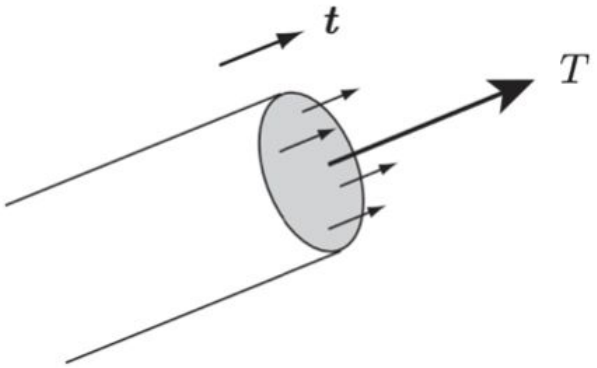
Find: Draw the free-body and kinetic diagrams of the block when s=0.4 m.

Inextensible strings

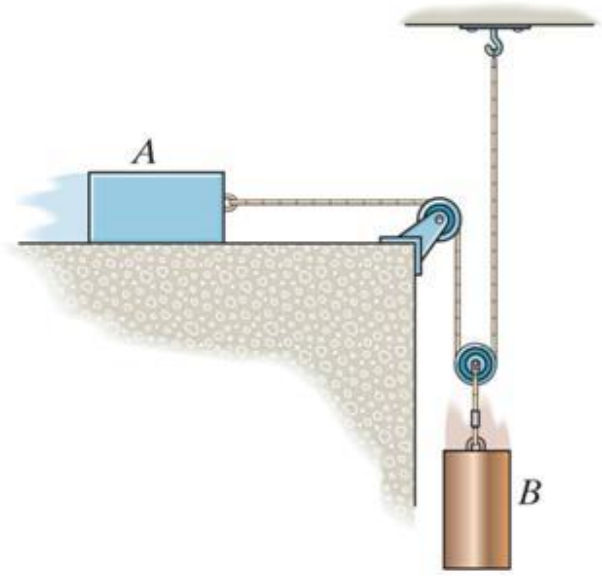
The inextensible string is an idealisation of real cords and ropes in that it is infinitely thin, has no bending stiffness, and is inextensible.

Its mass is negligible.

The only force that one part of the string exerts on another is the tension T in the string, which acts parallel to the tangent vector $\boldsymbol{t}$ to the string at each point.



Example 2



Given: The block and cylinder have a mass of m. The coefficient of kinetic friction at all surfaces of contact is μ. Block A is moving to the right.

Find: Draw the free-body and kinetic diagrams of each block.

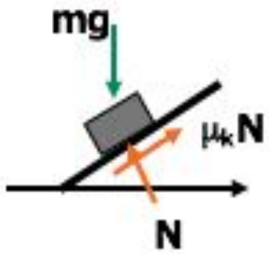
Quiz

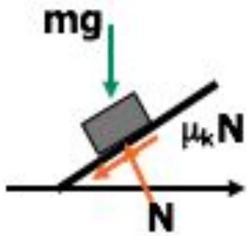
Internal forces are not included in an equation of motion analysis because the internal forces are_____

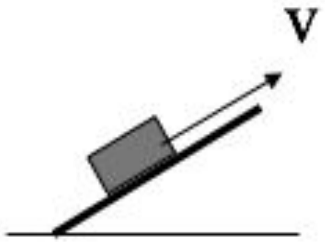
A) equal to zero.
B) equal and opposite and do not affect the calculations.
C) negligibly small.
D) not important.

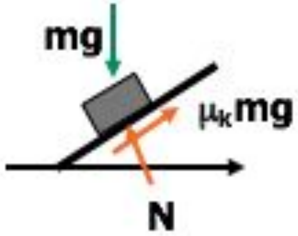
Quiz

The block (mass = m) is moving upward with a speed v. Draw the FBD if the kinetic friction coefficient is μ_k.

A) 

B) 

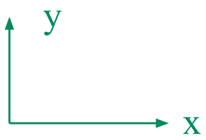


C) 

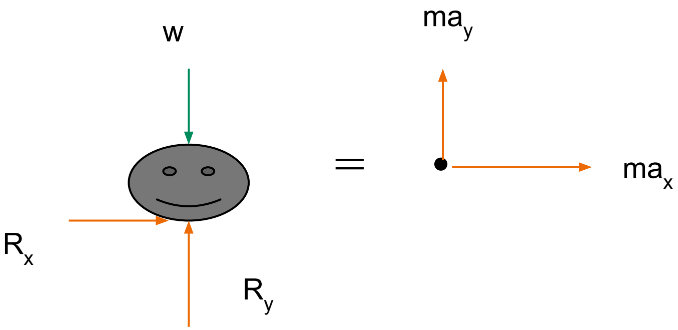
D) None of the above.

Quiz

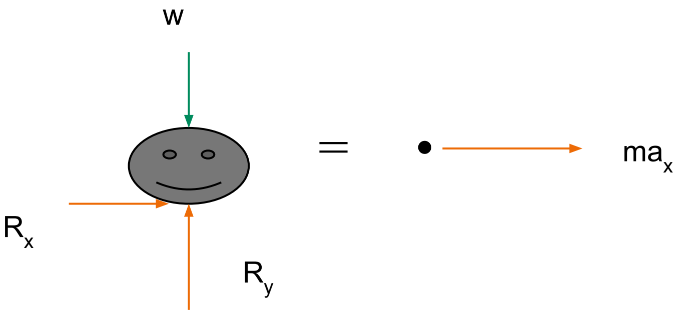
Packaging for oranges is tested using a machine that exerts $a_y = 20 \text{ m/s}^2$ and $a_x = 3 \text{ m/s}^2$, simultaneously. Select the correct FBD and kinetic diagram for this condition.



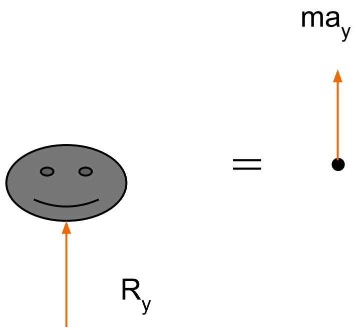
A)



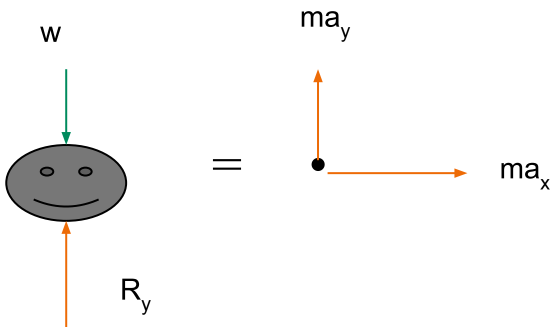
B)



C)

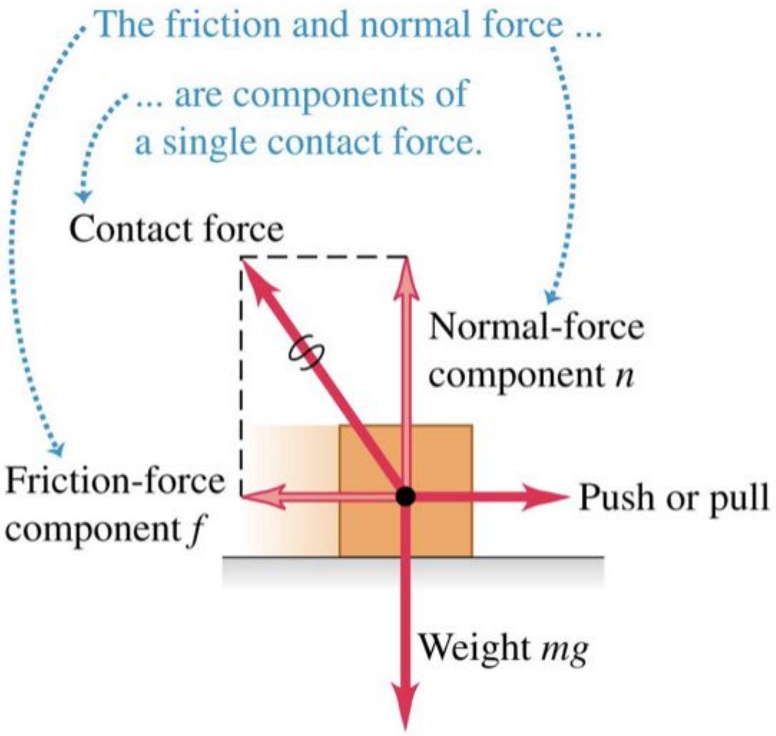


D)



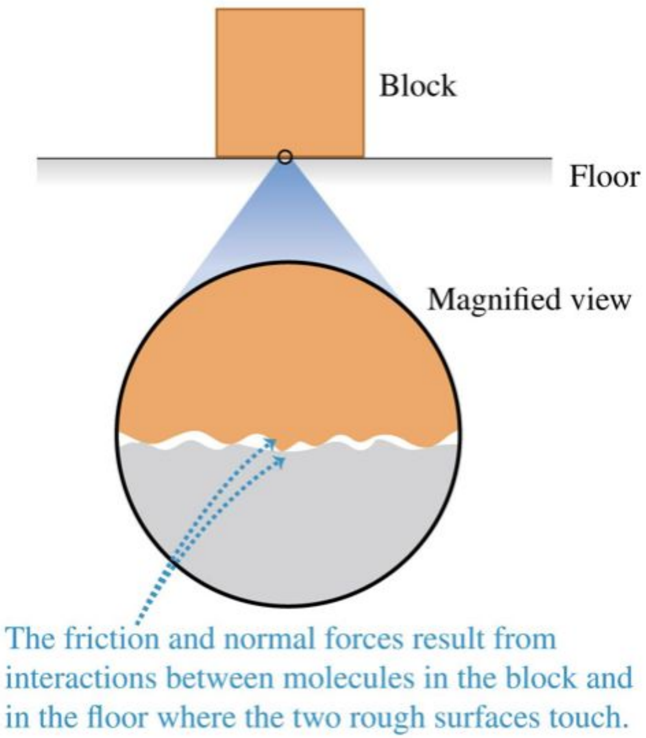
Frictional forces

When a body rests or slides on a surface, the friction force is parallel to the surface.



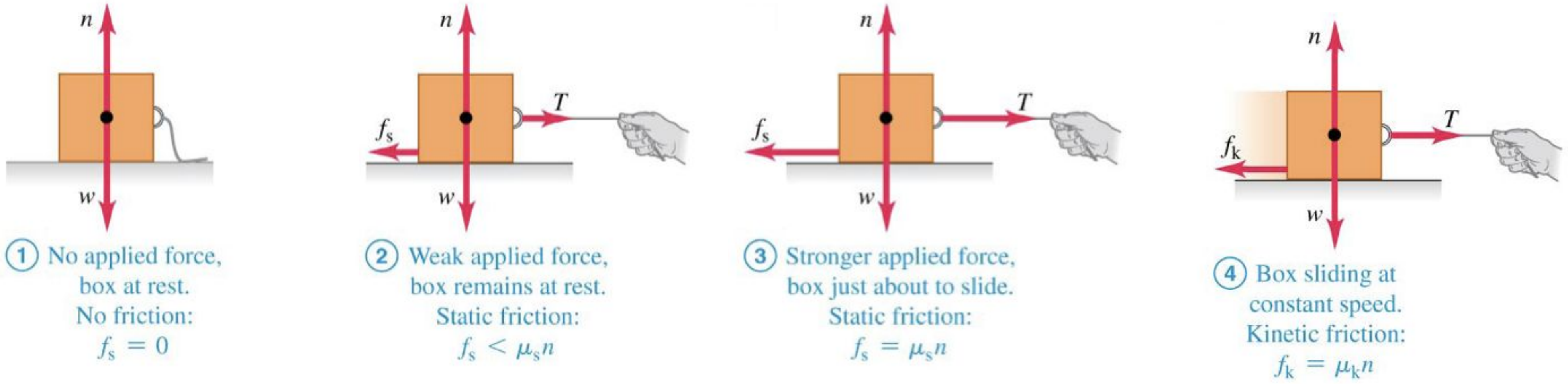
Frictional forces

Friction between two surfaces arises from interactions between molecules on the surfaces.

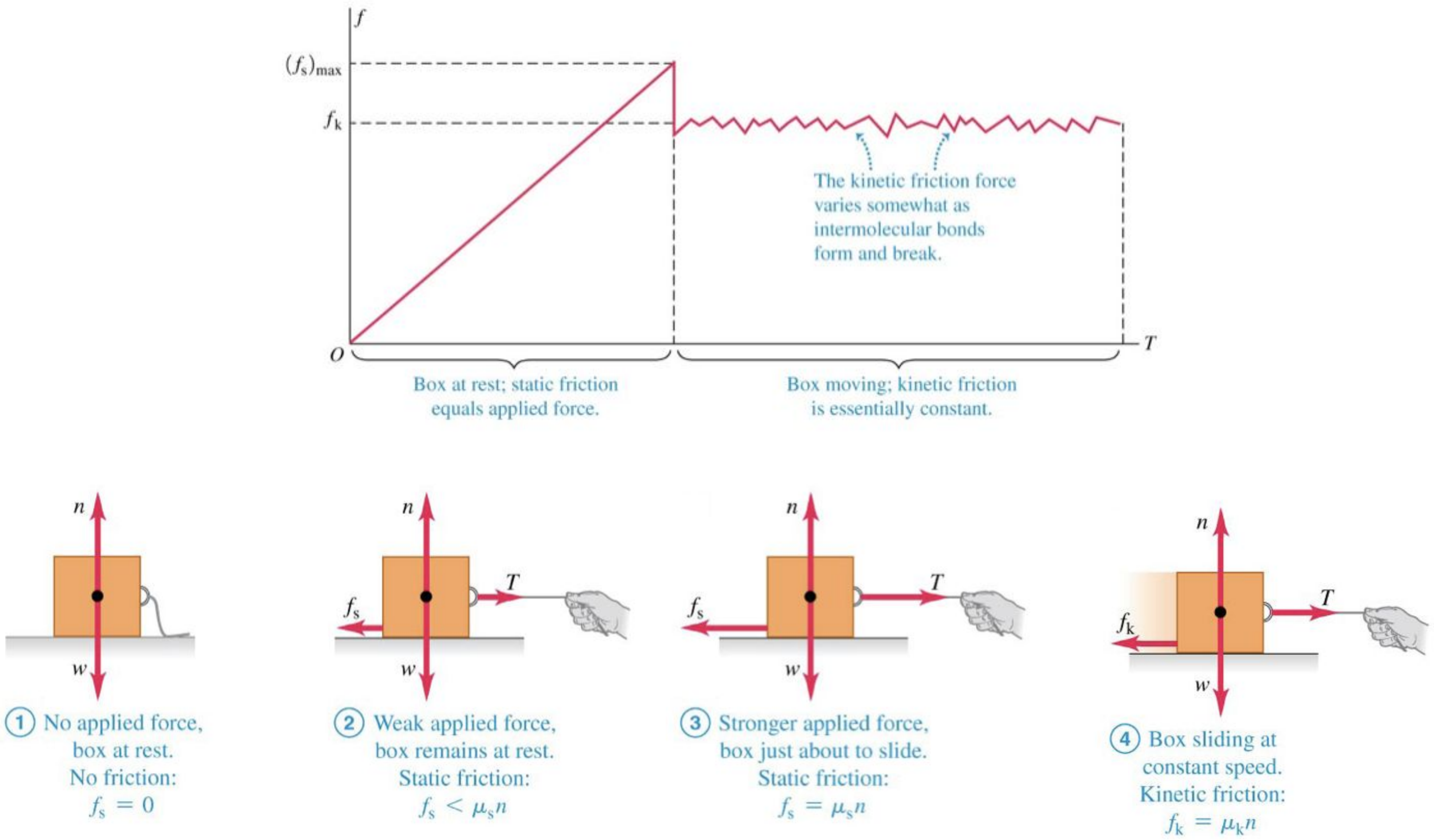


Kinetic and static friction

- Kinetic friction acts when a body slides over a surface.
- The kinetic friction force is $f_k = \mu_k n$.
- Static friction acts when there is no relative motion between bodies.
- The static friction force can vary between zero and its maximum value: $f_s \le \mu_s n$.



Kinetic and static friction

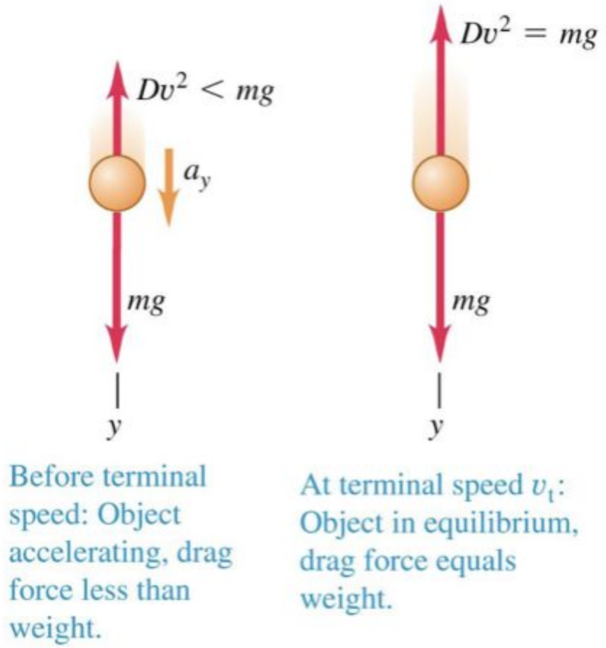


Air/water resistance

- When a body moves through a fluid such as air or water, the fluid exerts forces on the body.

- Because the body must push the fluid out of the way, by the Third Law, the fluid must then exert equal and opposite forces on the body.

- The expression of the resistance force depends on the velocity of the object.
  - If v is large, $F_d = D\, v^2$
  - If v is small, $F_d = D'\, v$



Quiz

The drag force F_d on a sphere moving slowly through a viscous fluid depends on the viscosity of the fluid η (with units kg/(m·s)), the radius R, and the speed v. Which of the following quantities is F_d?

A. $6\pi\eta R/v$

B. $6\pi\eta/Rv$

C. $6\pi\eta Rv$

D. $6\pi\eta R^2v$

Tenet (2020)

The film does not discuss the possibility of time traveling; rather, it depends on an extrapolation of inverting the internal clock of an object while the past was still going forward.

The film also suggests that anyone making the trek back to the past needs protection (such as a mask) to prevent any harm done to themselves.

Questions

What did you notice in the scene? Anything unusual caught your eyes?

Without discussing the plot, why does the flow of time seem strange in the clip?

Do the laws of classical mechanics allow this to happen? Can you think of any related concepts? (try not to meddle with thermodynamics and entropy)

Summary

- The Three Laws of Motion
- Inertial Frame
- Mass and weight
- Friction
- Resistance
- Time reversal symmetry